Supplementary Material: NLP-Augmented Multimodal Attention Fusion for Immunotherapy Response Prediction in Non-Small Cell Lung Cancer

S1 Supplementary Methods

Supplementary Note S1: NLP Prompt Template

Each patient's 13 clinical variables were converted into a short English patient description of ten sentences using the following template (placeholders shown in `[brackets]`):

> *"The patient is a [AGE]-year-old. Smoking history in pack-years is [PACK-YEARS]. ECOG performance status is [ECOG]. Blood albumin concentration is [ALBUMIN]. Derived neutrophil-to-lymphocyte ratio (dNLR) is [DNLR]. Initial tumor burden stands at [TUMOR-BURDEN]. The patient is diagnosed with [HISTOLOGY] at the [PRIMARY-SITE] site. Metastatic status: [BRAIN-METS] brain metastasis and [LIVER-METS] liver metastasis. Currently undergoing therapy line [THERAPY-LINE], utilizing [THERAPY-TYPE]. The patient [PRIOR-PD-L1] prior PD-L1 immunotherapy."*

The 13 variables are age, smoking pack-years, ECOG performance status, albumin, dNLR, tumor burden, histology, primary site, brain metastasis, liver metastasis, therapy line, combination therapy and prior PD-L1 therapy. Binary variables were expanded into descriptive phrases rather than coded values: [HISTOLOGY] $\in$ {adenocarcinoma, non-adenocarcinoma}; [PRIMARY-SITE] $\in$ {lung, extra-pulmonary}; [BRAIN-METS] and [LIVER-METS] $\in$ {with, without}; [THERAPY-TYPE] $\in$ {combination therapy, monotherapy}; [PRIOR-PD-L1] $\in$ {received, did not receive}. Age and therapy line were inserted as integers; albumin, dNLR and tumor burden with two decimal places; pack-years and ECOG as stored in the source table. Missing values were set to zero before sentence construction (for binary variables, zero maps to the negative phrase).

A complete example generated for one discovery-cohort patient:

> *"The patient is a 69-year-old. Smoking history in pack-years is 48.0. ECOG performance status is 1.0. Blood albumin concentration is 4.00. Derived neutrophil-to-lymphocyte ratio (dNLR) is 2.40. Initial tumor burden stands at 52.00. The patient is diagnosed with adenocarcinoma at the lung site. Metastatic status:*

without brain metastasis and without liver metastasis. Currently undergoing therapy line 2, utilizing combination therapy. The patient received prior PD-L1 immunotherapy."

Across the discovery cohort, sentences contained 65–68 words and 129–135 tokens under the all-MiniLM-L6-v2 tokenizer (including special tokens), below the encoder's 256-token input limit, so no sentence was truncated.

Supplementary Note S2: Domain-Specific Encoder (BioClinBERT)

We additionally evaluated `Bio_ClinicalBERT` [1], a BERT model pretrained via masked language modelling (MLM) on MIMIC-III clinical notes, as an alternative to the general-purpose `all-MiniLM-L6-v2` sentence encoder. Embeddings were obtained by attention-masked mean pooling over the final hidden layer (768 dimensions) and PCA-reduced to 16 dimensions (BioClinBERT-PCA16), mirroring NLP-PCA16. The input text for this encoder was a shorter prompt variant focused on immunotherapy-related risk factors (age, ECOG, albumin, dNLR, therapy line, combination therapy, histology, brain and liver metastasis, followed by a list of adverse risk factors such as hypoalbuminemia or elevated dNLR), rather than the template of Supplementary Note S1.

BioClinBERT-PCA16 was evaluated in the IHC-A arm only, where it reached AUC = 0.767 [0.701–0.833], below MiniLM-based NLP-PCA16 (AUC = 0.784 [0.719–0.848]; ΔAUC = -0.017) and on par with numeric labs (AUC = 0.768). Because the two encoders were given different prompt variants, this comparison reflects the encoder and the prompt jointly and should not be read as a controlled comparison of the encoders alone. A plausible contributing factor on the encoder side is the pretraining objective: BioClinBERT's MLM objective optimises the prediction of masked tokens rather than sentence-level embeddings in which semantically similar sentences lie close together, and mean pooling of such token embeddings is known to yield anisotropic embedding spaces, whereas all-MiniLM-L6-v2 was trained with a contrastive sentence-similarity objective.

Supplementary Note S3: Power Analysis for AUC Comparisons

To contextualise the overlapping DeLong confidence intervals reported for ΔAUC values of $+0.017$ to $+0.030$, we performed a post-hoc power analysis. With $n = 247$ (62 responders / 185 non-responders, prevalence $\approx$25%), the Hanley–McNeil 95% confidence interval half-width for a single AUC near 0.78–0.81 is approximately ± 0.07, consistent with the observed half-widths of ± 0.060–0.069 in Tables 5–6 of the main text.

For the difference between two correlated AUCs estimated on the same patients, assuming a between-method correlation of $\rho \approx 0.9$ (both models share all non-clinical modalities), $\mathrm{SE}_0(\Delta) = \sqrt{2(1-\rho)}\,\mathrm{SE}(\mathrm{AUC}) \approx 0.014$–$0.016$ at $n_0 = 247$. Since the standard error scales with $1/\sqrt{n}$, the sample size required to detect a true difference Δ with 80% power ($\alpha = 0.05$, two-sided; $z_{\alpha/2} = 1.96$, $z_\beta = 0.84$) is

$$n \;\geq\; n_0 \left(\frac{(z_{\alpha/2} + z_\beta)\,\mathrm{SE}_0(\Delta)}{\Delta} \right)^2 ,$$

which gives $n \approx 950$–1,250 patients for $\Delta\text{AUC} = 0.02$ and $n \approx 420$–550 for $\Delta\text{AUC} = 0.03$. The current discovery cohort ($n = 247$) is therefore underpowered by a factor of approximately 2–5$\times$ for the AUC differences observed. This calculation concerns the binary-response AUC comparison only and does not address the power of the survival analyses.

S2 Supplementary Figures

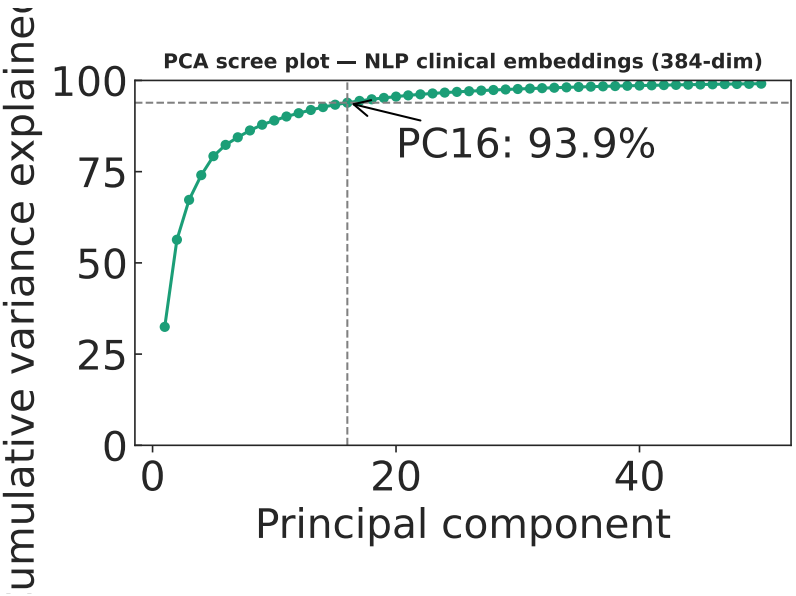


Figure S1: PCA scree plot for the 384-dimensional NLP sentence embeddings. The x-axis shows the principal component number (1–50) and the y-axis shows cumulative variance explained (%). The first 16 components (vertical dashed line, used as "NLP-PCA16") jointly explain ≈94% of total variance.

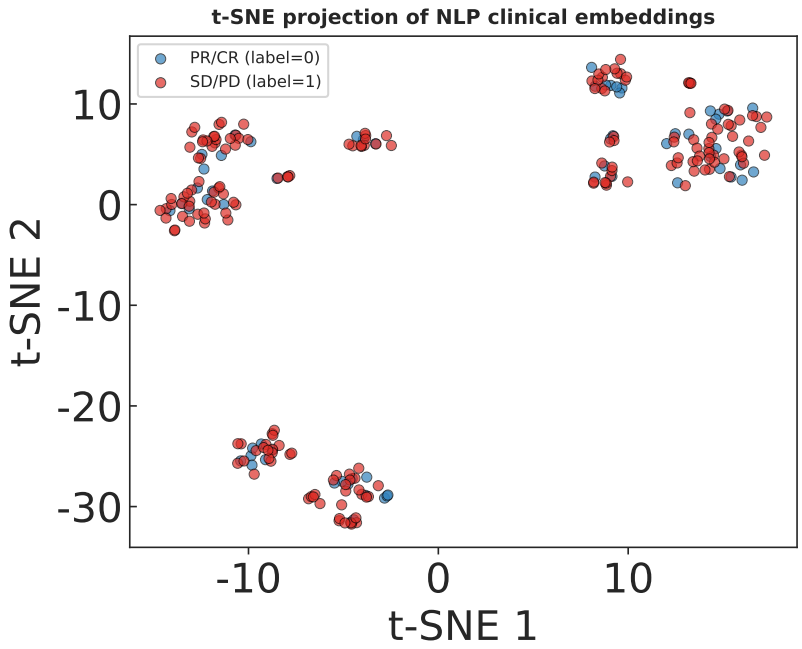


Figure S2: Two-dimensional t-SNE projection of the 384-dimensional NLP clinical embeddings for all $n = 247$ discovery-cohort patients, coloured by binary outcome (blue: PR/CR, label = 0; red: SD/PD, label = 1). The substantial overlap between classes is consistent with the embeddings encoding general clinical/prognostic similarity rather than a strong linear separation of binary treatment response, in agreement with the survival-dominant signal described in the Survival Analysis section of the main text.

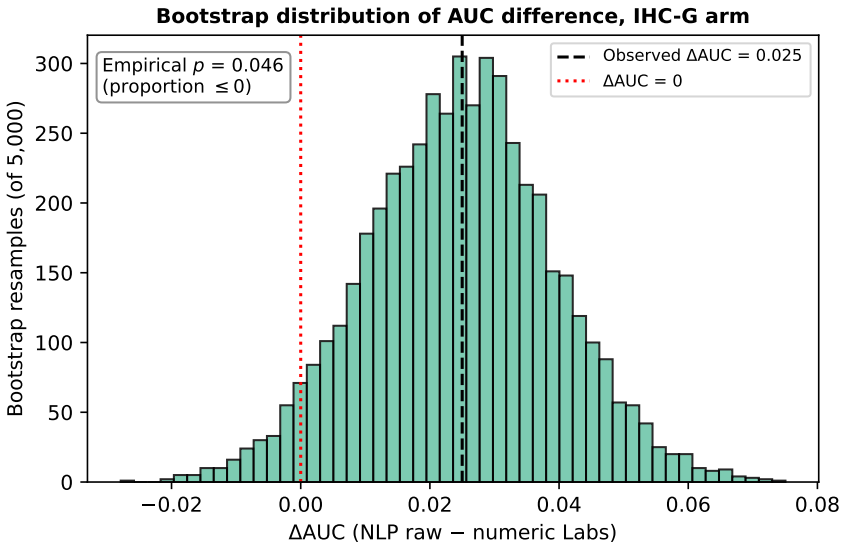


Figure S3: Bootstrap distribution of ΔAUC (NLP raw − numeric Labs, IHC-G arm) across 5,000 paired bootstrap resamples of the discovery cohort. The vertical dashed line marks the observed $\Delta\text{AUC} = 0.026$. The proportion of resamples with $\Delta\text{AUC} \leq 0$ provides an empirical bootstrap p-value, complementing the DeLong confidence intervals reported in Table 6 of the main text.

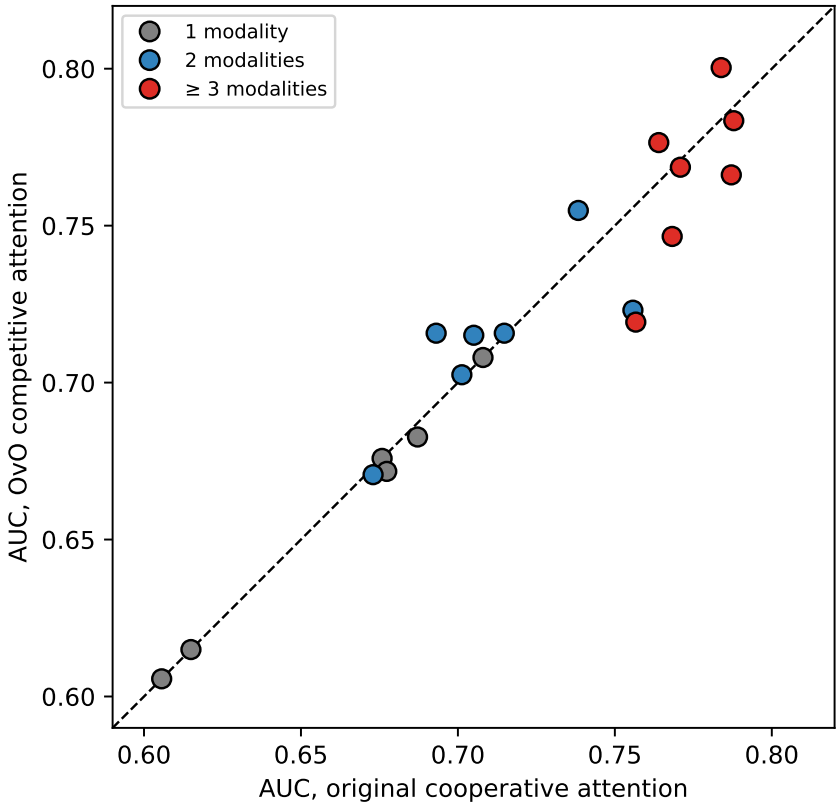


Figure S4: Scatter plot comparing AUC under the original cooperative attention (x-axis) versus OvO competitive attention (y-axis) across all 20 tested modality configurations. The diagonal line indicates equal performance. Points are coloured by the number of fused modalities (grey: 1 modality, where OvO reduces to the original formulation; blue: 2 modalities; red: $\geq$3 modalities). Points above the diagonal indicate configurations where OvO attention improved AUC.

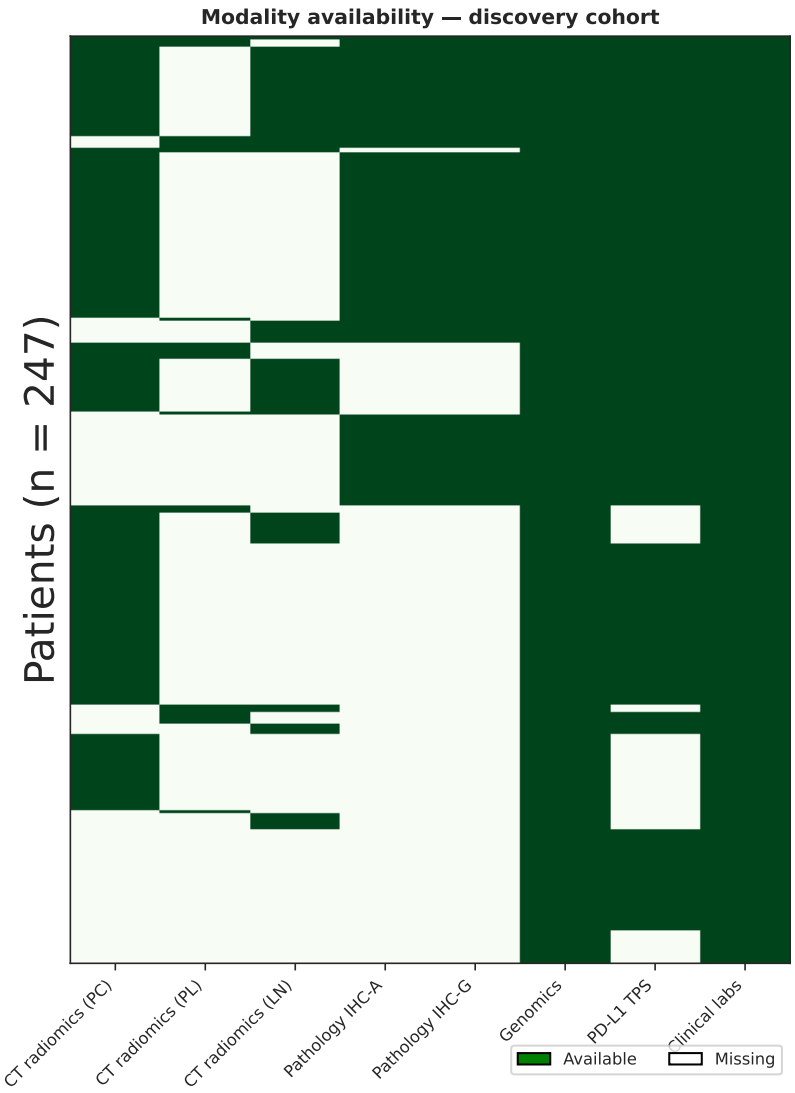


Figure S5: Modality availability heatmap for the discovery cohort ($n = 247$ patients $\times$ 8 modalities/sub-modalities: CT radiomics PC/PL/LN, pathology IHC-A, pathology IHC-G, genomics, PD-L1 TPS, and clinical labs). Green indicates the modality is available for a given patient; white indicates missing data, handled via the binary missingness mask described in the Methods section of the main text.

S3 Supplementary Tables

Table S1: Modality availability in the discovery cohort ($n = 247$). Counts are taken from the binary missingness mask actually used by the fusion model (`modality_MASK`), so they are the number of patients contributing that modality during training. A patient may contribute lesions at more than one anatomical site; the "any site" row is the union used for the CT radiomics entry of Table 1 of the main text.

Modality	**Patients available**	**Coverage (%)**
CT radiomics (PC lesion)	163	66.0
CT radiomics (PL lesion)	21	8.5
CT radiomics (LN lesion)	67	27.1
CT radiomics (any site)	187	75.7
Pathology IHC-A	105	42.5
Pathology IHC-G (GLCM)	105	42.5
Genomics (MSK-IMPACT)	247	100.0
PD-L1 TPS	201	81.4
Clinical labs (13 variables)	247	100.0

Table S2: Feature counts per modality before and after feature selection. Only the CT radiomics modalities pass through the ICC-based robustness filter (ICC > 0.15), outlier filter (z-score < 6) and elastic-net L1 selection described in the Methods section of the main text; all other modalities enter the model at full dimensionality. Because the selection is refitted inside every training fold (the held-out fold is excluded before fitting, to avoid selection leakage), the number of retained radiomics features varies by fold and is reported as the median across the 10 folds with the observed range in brackets.

Modality	**Raw features**	**After selection**	**Retention (%)**
CT radiomics (PC lesion)	1688	35 [28–45]	2.1
CT radiomics (PL lesion)	1688	9 [3–20]	0.5
CT radiomics (LN lesion)	1688	25 [17–32]	1.5
Pathology IHC-A	18	18	100.0
Pathology IHC-G (GLCM)	150	150	100.0
Genomics	11	11	100.0
PD-L1 TPS	1	1	100.0
Clinical labs	13	13	100.0
NLP raw (MiniLM-L6-v2)	384	384 (no_scale)	100.0
NLP-PCA16	16	16 (no_scale)	100.0

Table S3: Single-partition (single 10-fold split, natural patient order, the protocol used in the exploratory screen of the main text) AUC across the same 21 configurations and three model variants as Table 4 of the main text, which reports the corresponding repeated-seed values. As discussed in the OvO Attention section of the main text, single-partition values are more favourable to whichever model happens to fit the specific fold assignment and should not be interpreted as evidence of a reproducible advantage; they are reported here only for comparability with prior single-partition analyses.

#	Configuration	DyAM	OvO	OvO+NLP
1	TMB	0.615	0.615	0.610
2	PDL1	0.708	0.708	0.678
3	IHC-A	0.606	0.606	0.660
4	Gen	0.676	0.676	0.655
5	Rad	0.693	0.679	0.706
6	Rad-LU	0.678	0.671	0.691
7	TMB+PDL1	0.636	0.699	0.698
8	PDL1+Gen (BM4)	0.693	0.716	0.737
9	Rad+IHC-A	0.688	0.681	0.746
10	Rad+IHC-G	0.710	0.700	0.775
11	Rad+Gen (BM3)	0.755	0.751	0.755
12	IHC-A+Gen	0.715	0.716	0.723
13	IHC-G+Gen	0.756	0.723	0.756
14	Rad+IHC-A+Gen	0.772	0.736	0.801
15	Rad+IHC-G+Gen	0.783	0.769	0.803
16	Rad+IHC-A+MutAmp	0.782	0.767	0.768
17	Rad+IHC-A+Gen+PDL1	0.796	0.767	0.798
18	Rad+IHC-G+Gen+PDL1 (BM1)	0.798	0.797	0.812
19	Rad+IHC-A+Gen+TMB+PDL1	0.789	0.791	0.813
20	Rad+IHC-A+Gen+PDL1+Labs	0.765	0.762	0.798
21	Rad+IHC-G+Gen+PDL1+Labs (BM2)	0.791	0.778	0.812

References

[1] Emily Alsentzer, John R. Murphy, Willie Boag, et al. Publicly available clinical BERT embeddings. In *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, pages 72–78, 2019.